# Euclidean Transformations (Rotation)

Thursday, April 16, 2026 12:06 PM

$f(x,y) = x^2 - 2y + 5$

**Translation** vector $\begin{pmatrix} -3 \\ 5 \end{pmatrix}$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

$x' = x - 3 \quad , \quad y' = y + 5$

$x' + 3 = x \quad ; \quad y' - 5 = y$

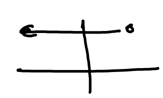
$$f(x', y') = (x'+3)^2 - 2(y'-5) + 5$$
$$= x'^2 + 6x' + 9 - 2y' + 10 + 5$$
$$= x'^2 - 2y' + 6x' + 24$$

**Reflection**

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

x-axis $(y=0)$ $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$(x,y) \to (x,-y)$



y-axis $(x=0)$ $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

$y = x$, $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ✓

$T(e_1) = T(1,0) = (0,1)$

$T(e_2) = T(0,1) = (1,0)$

$y = -x$ $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

$2x + 3y = 5$ in $y = -x$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$
$$= \begin{pmatrix} -y \\ -x \end{pmatrix}$$

$x' = -y \quad ; \quad y' = -x$

$-x' = y \quad , \quad -y' = x$

$2(-y') + 3(-x') = 5$

$-2y' - 3x' = 5 \Rightarrow 3x' + 2y' = -5$

# Rotation

* fixed Point C
* Moving Point (object) (P)
* angle + direction $\theta$
  Anticlockwise (ACW)
  clockwise, C-W)
* New point (Image) $P'$

Condition $\boxed{CP = CP'}$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix} \qquad A = 2\times 2$$

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

A.C.W Rotational matrix about Centre of Rotation Origin

$$\cos(-\theta) = \cos\theta$$

$$\sin(-\theta) = -\sin\theta$$

$P(-1, 2)$; Rotate $90^\circ$ A.C.W. Through origin

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos 90 & -\sin 90 \\ \sin 90 & \cos 90 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

$P(-3, -2)$ $90^\circ$ clockwise

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos(-90) & -\sin(-90) \\ \sin(-90) & \cos(-90) \end{pmatrix} \begin{pmatrix} -3 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -3 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

① centre of circle A $(-2, 4)$

Apply following Transformat $T\,M\,\underline{R\,(-2, 4)}$

$T =$ translation in $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$M$: Reflection in line $y = x$

$R$ = Rotation in $180^\circ$, $(0,0)$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos 180 & -\sin 180 \\ \sin 180 & \cos 180 \end{pmatrix} \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

$$TM\begin{pmatrix} 2 \\ -4 \end{pmatrix} \qquad = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$$

$$\begin{pmatrix} x'' \\ y'' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -4 \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \end{pmatrix} \longrightarrow T\begin{pmatrix} -4 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x''' \\ y''' \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \begin{pmatrix} -4 \\ 2 \end{pmatrix}$$

$$TMR\begin{pmatrix} -2 \\ 4 \end{pmatrix} = \underline{\begin{pmatrix} -6 \\ 3 \end{pmatrix}}$$

$$R^2\begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

$$= R\,R\begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

**Q** Rotate line $2x - 3y = 8$, Through $270^\circ$ clockwise

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos(-270) & -\sin(-270) \\ \sin(-270) & \cos 270 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 \\ \cdots & \cdots \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

$$x' = -y \Rightarrow y = -x'$$
$$y' = x$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$$

$$2y' - 3(-x') = 8 \qquad 3x' + 2y' = 8$$

Rotate $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$

$\theta = 30^\circ$, anticlockwise

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos(30) & -\sin(30) \\ \sin(30) & \cos 30 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$x' = \frac{\sqrt{3}}{2}x - \frac{1}{2}y \Rightarrow 2x' = \sqrt{3}x - y$$
$$y' = \frac{1}{2}x + \frac{\sqrt{3}}{2}y \Rightarrow 2y' = x + \sqrt{3}y$$

$$2x' = \sqrt{3}x - y \Rightarrow 2\sqrt{3}x' = 3x - \cancel{\sqrt{3}y}$$
$$2y' = x + \cancel{\sqrt{3}y}$$
$$2y' = x + \sqrt{3}y$$

$$2\sqrt{3}x' + 2y' = 4x$$

$$\boxed{\frac{\sqrt{3}}{2}x' + \frac{1}{2}y' = x}$$

$$2\sqrt{3}y' = \cancel{\sqrt{3}x} + 3y$$
$$2x' = \cancel{\sqrt{3}x} - y$$

$$2\sqrt{3}y' - 2x' = 4y$$

$$\boxed{-\frac{1}{2}x' + \frac{\sqrt{3}}{2}y' = y}$$

if $2x + 4y = 20$

$$2\left(\frac{\sqrt{3}}{2}x' + \frac{1}{2}y'\right) + 4\left(-\frac{1}{2}x' + \frac{\sqrt{3}}{2}y'\right) = 20$$

$$\sqrt{3}\,x' + y' - 2x' + 2\sqrt{3}\,y' = 20$$

$$(\sqrt{3}-2)x' + (1+2\sqrt{3})y' = 20$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1 \qquad = \qquad \frac{\left(\frac{\sqrt{3}}{2}x' + \frac{1}{2}y'\right)^2}{4} + \frac{\left(-\frac{1}{2}x' + \frac{\sqrt{3}}{2}y'\right)^2}{9} = 1$$